

Nikolai Daelemans

nikolai@daelemans-experts.eu | <https://www.linkedin.com/in/nikolai-daelemans/> | www.nikolaidaelemans.com

SUMMARY

Software Engineer & CTO with experience in AI automation, cloud infrastructure, and full-stack development. Currently building AvantifAI while leading AI platform engineering at KBC. Seeking hybrid/remote mid-level engineering roles in AI or infrastructure.

SKILLS

Tech Stack: Python, Java, C#, JavaScript/TypeScript, React, Angular, Vue, SQL, Postgres
AI/ML: PyTorch, TensorFlow, LLMs, RAG, Vector Databases, NLP, Prompt Engineering, ML Engineering
Sys/Infra: Azure, AWS, Docker, Kubernetes, CI/CD, REST API, Microservices, DevOps, Kafka, Redis, Elastic, Jenkins, Git, PowerBI, Fabric, Kibana, PySpark, Pandas, Agile/Scrum, Jira, Firewall/Zero Trust, IaC
Languages: Dutch (C2), English (C2), French (B2), German (A2), Luxembourgish (A2)
Interests: AI, Programming, Trading, Finance, Fitness, Tennis, Running, Reading, Travelling, KiteSurfing
Certifications: Microsoft Certified: Azure AI Fundamentals (2025), Datacamp: Efficient AI model training with Pytorch, Introduction to PowerBI

EXPERIENCE

CTO & Co-founder – AvantifAI (www.avantifai.com)

Based in Europe (March 2026 - present)

- Running an AI consultancy company designing solutions for mid-to-large companies. Saving multiple hours per employee per week by facing real pain points.
- Developed & designed DocAI, with Agentic AI/Browser Extension/Version Control/Multi-tenancy/App Security/Rate limiting/Vectorizing/db backups/HTTPS/Auth0/access management with the purpose of improving enterprise level documentation structures as an intelligent documentation manager.

Infrastructure Software Engineer (AI Platforms & Automation) – KBC

Leuven, Belgium (July 2025 - present)

- Built reusable **Infrastructure as Code (IaC)** frameworks to automate provisioning with AI assisted decision making across diverse infrastructure systems using Python and PowerShell.
- Automated the process of users requests for Compute (On-Prem Virtualization), Storage operations (VMware, SAN, NAS, XP8), Firewall (Rules, Zero Trust, IDS, IPS), Load Balancing (L4, L7), DDI (DNS), Draining tool (Netscaler, F5)
- Team lead** of KBC's draining tool, organizing sprints and managing the team with Jira in the agile methodology to achieve qualitative results.
- AI Innovation & Engineering** - Developing custom AI models and integrating Microsoft Copilot and Power Automate solutions to streamline internal banking processes. Troubleshooting recurring issues reported via internal support channels and automating resolutions to reduce human error.

Technology Consultant - PwC (Cloud & Data intern, Digital Transformation/Technology)

Brussels, Belgium (February 2025 - May 2025)

- Led the organization and technical preparation together with Microsoft of a hackathon for the NATO, coordinating multi-disciplinary teams and aligning deliverables with defense innovation goals, resulting in actionable MARCOM data solutions and strategic prototypes that informed NATO maritime safety protocols.
- Automated a time-intensive multi-excel workflow used by cross-functional teams, developing a Python script to ingest, process, and insert records-eliminating manual data entry and saving 20+ engineering hours per month.
- Preprocessed and cleaned large-scale defense datasets using Python (Pandas, NumPy), enabling accurate analytics and feature engineering.
- Developed dynamic visualizations and dashboards with Power BI and Plotly to highlight key patterns and threats from complex data structures, improving interpretability and decision-making for NATO analysts and leadership.

Software Engineer – Tejo

Hybrid (October 2024 - December 2024)

- Designed and built a full event payment pipeline for a local non-profit — hosted frontend, Stripe/payment integration, automated confirmation emails with generated QR codes for event entry. Handled 100% digital payment processing with zero manual intervention on event day.

Software Engineer (Product Owner) - Belfius

Belgium (September 2024-December 2024)

- Coordinated go-to-market technical readiness for an internal product modernization at Belfius, aligning engineering deliverables with business launch milestones and stakeholder requirements improving system performance by 50%.
- Designed and deployed scalable microservices using .NET Core and REST APIs, while contributing to Angular and React-based front-end enhancements to improve UX for internal users.

Lead IT & Analytics - Emergent

Leuven, Belgium (March 2024-March 2025)

- Led the digital strategy and analytics initiatives for a student-run AI & Data Science organization, partnering with top-tier companies such as McKinsey, KBC, AWS, Deloitte, DataCamp and more to host workshops, talks, and hands-on projects; enhanced community engagement by 30% through scalable web solutions (WordPress/Elementor) and analytics-driven content planning (Python, PySpark).

Software Engineer (App Development) – SunnySet

Hybrid (January 2024 - September 2024)

- Built and deployed a cross-platform mobile app using Flutter and Java to deliver real-time safety alerts and local news for tourists, enhancing urban navigation and user security in unfamiliar cities.
- Led core contributions across app logic, UI, and hosting, integrating CI/CD pipelines and managing web deployment to ensure smooth performance across devices.

Software Engineer Intern - Think-E

Leuven, Belgium (July 2023 - September 2023)

- Developed energy-efficient smart home applications by integrating IoT devices with real-time monitoring systems across residential complexes; leveraged C++, JavaScript, and SQL to connect sensors via Raspberry Pi, optimizing sustainability data collection and automation.
- Collaborated on full-stack solutions to improve energy tracking dashboards, incorporating responsive UI (HTML/MUI/CSS/JS) and GitHub-based version control to support scalable, maintainable systems.

EDUCATION:

B.Sc. Applied Computer Science — UCLL, Belgium (2025)

PROJECTS

2025 -AI & Infrastructure Learning Platform: Designing and building a learning platform. Focused on teaching infrastructure and AI theory to designing personal models and hands-on experience with IoT devices. Implemented a user feedback form for improvements

2024 - AI Questionnaire Generator and Interview Analysis System: Built using Python, NLP, and LLMs to automate interview analysis.

2023 - Campus Assistant Chatbot (<https://github.com/NikolaiDoesCode/CampusAI>) Hybrid chatbot combining a custom-trained PyTorch intent classifier with DialoGPT fallback. Bag-of-words preprocessing pipeline, 5-layer MLP, deployed via Flask with a responsive frontend.

2022 (Refractored in 2026) - FaceDetectorCNN (<https://github.com/NikolaiDoesCode/Face-Detection>): Built a real-time face detection system from scratch using a custom CNN in PyTorch. Trained on 26,000 samples (LFW + generated negatives), achieving 99.89% validation accuracy and AUC-ROC of 1.0. Implemented sliding window detection, image pyramid, and non-maximum suppression from scratch.